

Keiichi Hashimoto

PhD Student, Department of Earth and Planetary Science, Graduate School of Science, The University of Tokyo

Education

- **Ph.D. in Science (Earth and Planetary Science)**, The University of Tokyo (Apr 2024 – Present)
- **M.Sc. in Science (Earth and Planetary Science)**, The University of Tokyo (Apr 2022 – Mar 2024)
- **B.Sc. in Science (Earth and Planetary Physics)**, The University of Tokyo (Apr 2018 – Mar 2022)

Professional Experience

- **Temporary Research Assistant**, Japan Agency for Marine-Earth Science and Technology (JAMSTEC) (Aug 2025 – Mar 2026)
- **Technical Assistant**, Graduate School of Science, The University of Tokyo (Apr 2025 – Mar 2026)
- **Research Student**, Japan Agency for Marine-Earth Science and Technology (JAMSTEC) (Jan 2022 – Present)

Publications

First Author (Under Review)

- K. Hashimoto, T. Hajima, H. Tatebe, T. Kataoka, and H. Miura: An ENSO amplitude enhancement by the dynamic simulation of LAI in the Earth System Model MIROC-ES2L. *J. Clim.*, in review.

Co-Author (Accepted/Published)

- O'Donnell, G.L., A.A. Wing, L.G. Silvers, K.A. Reed, P.A. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. M. Hannah, K. Hashimoto, P.G. Hill, G. Leung, S. Matsugishi, S. Ortega, R. Roehrig, D.M. Romps, J. Shpund, A. Sokol, L. Tomassini, S. van den Heever, and C.-M. Wu: Characteristics of a multimodel ensemble of mock-Walker simulations. *J. Adv., Model Earth Syst.*, accepted.

Presentations

First Author

- K. Hashimoto, T. Hajima, and H. Miura: Development of a Unified Photosynthesis-Hydraulics Module for MIROC-ES2L. *JpGU-AGU Joint Meeting 2026*, Makuhari Messe, Chiba, 24 – 29 May 2026. [poster]
- K. Hashimoto, T. Hajima, H. Tatebe, T. Kataoka, and H. Miura: Land Components Control the ENSO Representation in the Earth System Model MIROC-ES2L. *3rd International Workshop on Interbasin Interaction and Climate Variability*, The University of Tokyo, Tokyo, Japan, 3 – 4 December 2025. [poster]
- 橋本恵一, 新田友子, 羽島知洋, 廣田渚郎, 三浦裕亮: 気候シミュレーションでの利用に向けた、ISS搭載型 LiDARによる全球樹冠高・樹冠底高格子データの整備. *日本気象学会2025年度秋季大会*, 福岡国際会議場, 福岡, 2025年11月4 – 8日. [poster]
- 橋本恵一: 最適性理論に基づく二葉式光合成・蒸散スキームの開発. *第11回マッデン・ジュリアン振動研究会*, 東京大学山中寮セミナーハウス, 山梨, 2025年9月25 – 26日. [oral]
- 橋本恵一, 羽島知洋, 三浦裕亮: 現実的な樹高がMIROC6による気候シミュレーションに与える影響. *2025年度GCM検討会*, 東北大学, 仙台, 2025年6月16 – 17日. [oral]

- K. Hashimoto, T. Hajima, and H. Miura: Impacts of Vegetation Canopy Height Distribution on Climate Simulations. *The 3rd UTokyo-NTU Workshop on Atmospheric Convection*, The University of Tokyo, Tokyo, 29 May 2025. [oral]
- K. Hashimoto, T. Hajima, and H. Miura: Sensitivity of the Coupled Atmosphere-Ocean System to Vegetation Height Distribution in the MIROC Climate Model. *JpGU 2025*, Makuhari Messe, Chiba, 25 – 30 May 2025. [poster]
- K. Hashimoto, T. Hajima, H. Tatebe, T. Kataoka, and H. Miura: Land components control the ENSO representation in the Earth System Model MIROC-ES2L. *EGU General Assembly 2025*, Vienna, Austria, 27 April – 2 May 2025. [poster]
- K. Hashimoto, T. Hajima, and H. Miura: Physical Effects of Terrestrial Vegetation Representation on Atmospheric Circulation in Earth System Models. *UTCCS Annual Workshop 2025*, The University of Tokyo, Tokyo, 6 March 2025. [poster]
- K. Hashimoto, T. Hajima, H. Tatebe, T. Kataoka, and H. Miura, The land components control the ENSO representation in the Earth System Model MIROC-ES2L. *The 4th Asian Conference on Meteorology (ACM) 2024*, Tsukuba International Congress Center, Tsukuba, 18 – 20 November 2024. [oral]
- 橋本恵一, 三浦裕亮: 地球システムモデルを用いた温暖化実験において動的植生がWalker循環の変化に与える影響. *日本気象学会2024年度秋季大会*, つくば国際会議場, つくば, 2024年11月12 – 15日. [oral]
- 橋本恵一: 地上設置型Rainscopeを用いた船上における降雨観測. *第10回マッデン・ジュリアン振動研究会*, 名古屋大学, 名古屋, 2024年10月29 – 30日. [oral]
- K. Hashimoto and H. Miura: Representation of terrestrial vegetation in Earth system models and its relationship with atmospheric circulation. *JpGU 2024*, Makuhari Messe, Chiba, 26 – 31 May 2024. [oral]
- K. Hashimoto: Representation of biogeophysical processes in South America in the earth system model MIROC-ES2L. *NTU-UTokyo Workshop on Atmospheric Convection*, National Taiwan University, Taipei, Taiwan, 2 – 4 March 2024. [oral]
- 橋本恵一, 羽島知洋, 建部洋晶, 片岡崇人, 三浦裕亮: 地球システムモデルMIROC-ES2LにおけるENSO表現と動的陸上植生の関係. *日本気象学会2023年度秋季大会*, 仙台国際センター, 2023年10月23 – 26日. [oral]
- 橋本恵一: 地球システムモデルMIROC-ES2LにおけるENSO表現と動的陸上植生の関係. *第9回マッデン・ジュリアン振動研究会*, 岡山大学, 岡山, 2023年9月28 – 29日. [oral]
- K. Hashimoto, T. Hajima, and H. Miura: Representation of biogeophysical processes in South America in the earth system model MIROC-ES2L. *JpGU 2023*, Makuhari Messe, Chiba, 21 – 26 May 2023. [oral]
- 橋本恵一: 地球システムモデルMIROC-ES2Lにおけるアマゾンの植生-大気相互作用の再現. *第8回マッデン・ジュリアン振動研究会*, お茶の水女子大学, 東京, 2023年1月18 – 19日. [oral]
- K. Hashimoto, C.-S. Hung, and H. Miura: Resolution and domain size dependence of cloud self-aggregation simulations in radiative-convective equilibrium systems. *JpGU 2022*, Makuhari Messe, Chiba, 22 May – 3 June 2022. [poster]

Co-Author

- O'Donnell, G.L., A.A. Wing, L. Silvers, K. Reed, P. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. Hannah, K. Hashimoto, P. Hill, G. Leung, S. Matsugishi, S. Ortega, R. Roehrig, D. Romps, J. Shpund, A. Sokol, L. Tomassini, S. van den Heever, C.-M. Wu: Organized convection in an ensemble of mock-Walker simulations. *37th Conference on Hurricanes and Tropical Meteorology*, San Diego, CA, USA, 30 March – 3 April, 2026. [oral]
- Wing, A.A., G.L. O'Donnell, L. Silvers, K. Reed, P. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. Hannah, K. Hashimoto, P. Hill, G. Leung, S. Matsugishi, S. Ortega, R. Roehrig, D. Romps, J. Shpund, A. Sokol, L. Tomassini, S. van den Heever, C.-M. Wu: Convective organization and climate feedbacks in mock-Walker simulations in RCEMIP-II. *106th AMS Annual Meeting*, Houston, TX, USA, 25-29 January 2026. [oral]

- O'Donnell, G.L., A.A. Wing, L. Silvers, K. Reed, P. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. Hannah, K. Hashimoto, P. Hill, S. Matsugishi, S. Ortega, R. Roehrig, D. Romps, J. Shpund, A. Sokol, L. Tomassini, T. Yanase, C.-M. Wu: Convective organization in mock-Walker simulations in RCEMIP-II. *Gordon Research Conference on Radiation and Climate*, Lewiston, ME, USA, 20-25 July 2025. [poster]
- O'Donnell, G.L., A.A. Wing, L. Silvers, K. Reed, P. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. Hannah, K. Hashimoto, P. Hill, S. Matsugishi, S. Ortega, R. Roehrig, D. Romps, J. Shpund, A. Sokol, L. Tomassini, T. Yanase, C.-M. Wu: Convective organization in mock-Walker simulations in RCEMIP-II. *Gordon Research Seminar on Radiation and Climate*, Lewiston, ME, USA, 19-20 July 2025. [poster]
- Wing, A.A., G.L. O'Donnell, L. Silvers, K. Reed, P. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fievet, B. Gasparini, W. Hannah, K. Hashimoto, P. Hill, S. Matsugishi, S. Ortega, R. Roehrig, D. Romps, J. Shpund, A. Sokol, L. Tomassini, T. Yanase, C.-M. Wu: Convective organization in mock-Walker simulations in RCEMIP-II. *2025 CFMIP/CloudSense Meeting*, Exeter, UK, 7-10 July 2025. [oral]

Awards

- **Matsuno Award**, Meteorological Society of Japan (MSJ) 2023 Autumn Meeting (Oct 2023)

Competitive Grants & Fellowships

- **JST Support for Pioneering Research Initiated by the Next Generation (SPRING) Program**, The University of Tokyo "Advanced Human Resource Development Leading Green Transformation (GX) (SPRING GX)" Project student (2024 – 2027)